

Task

Time limit: 1-2 day

Problem Statement

Develop a system to index any text and audio samples by creating embeddings that can represent both modalities in a common space. The goal is that related text and music should have similar embeddings, allowing for seamless retrieval and matching between them. To achieve this, we will explore various models and evaluate their performance on a dataset scraped from splice.com. The indexed data should be promptable, ie, user says **give me drum samples** the chatbot should return the drum samples from the db

We expect confusion matrices describing the difference in results from different models tried

Bonus: build without CLAP and use an audio only embedding model to align your text queries with the samples. Text query should match the audio embeddings stored in database (RAG).

Experimental Setup

Data Collection

- **Source:** splice.com
- **Classes:** Keys, Drums
- **Samples:** 20 from each class, please extract using browser extension

Your solution

Here is the comprehensive report content formatted for your Word document.

Project Report:

Cross-Modal Text-to-Audio Retrieval System

1. Executive Summary

This project implements an end-to-end Retrieval-Augmented Generation (RAG) system for audio samples, capable of indexing music files (Keys and Drums) and retrieving them via natural language text queries.

Key Achievement

The system was built following the bonus constraint: No pre-aligned models (CLAP) were used. Instead, we successfully trained a custom Neural Projector to align two disjoint vector spaces, Text Embeddings and Audio Embeddings, from scratch. This demonstrates a First Principles approach to cross-modal alignment.

2. Problem Statement

The objective was to develop a system to index text and audio samples by creating embeddings that represent both modalities in a common space. The system allows users to prompt naturally (e.g., "give me a heavy kick drum") and receive accurate audio samples from the database.

Experimental Setup

- Source: Splice.com
- Classes: Drums (Percussive, Transient-heavy) vs. Keys (Harmonic, Tonal)
- Constraint: Avoid pre-trained cross-modal models like CLAP. Use an audio-only model and align text queries to it manually.
- Database: ChromaDB for vector storage and retrieval.

3. Methodology: The Neural Bridge Architecture

Since we adhered to the constraint of not using CLAP, we treated the Text and Audio domains as two separate languages and built a translator (Neural Projector) between them.

System Pipeline

1. Audio Encoder: Extracts rich feature vectors from .wav files.
2. Text Encoder: Converts user queries into semantic vectors.
3. The Neural Bridge (MLP): A trained custom neural network that projects Text Embeddings into the Audio Embedding Space.
4. Dynamic Context Enrichment: To ensure robust training, filenames were expanded into rich captions during the training phase (e.g., "kick.wav" became "heavy thumping bass drum low frequency impact").

4. Model Iterations & Performance

We evaluated two distinct architectures to measure the impact of model size and domain-specific pre-training.

Model A: The Baseline (Lightweight)

Architecture

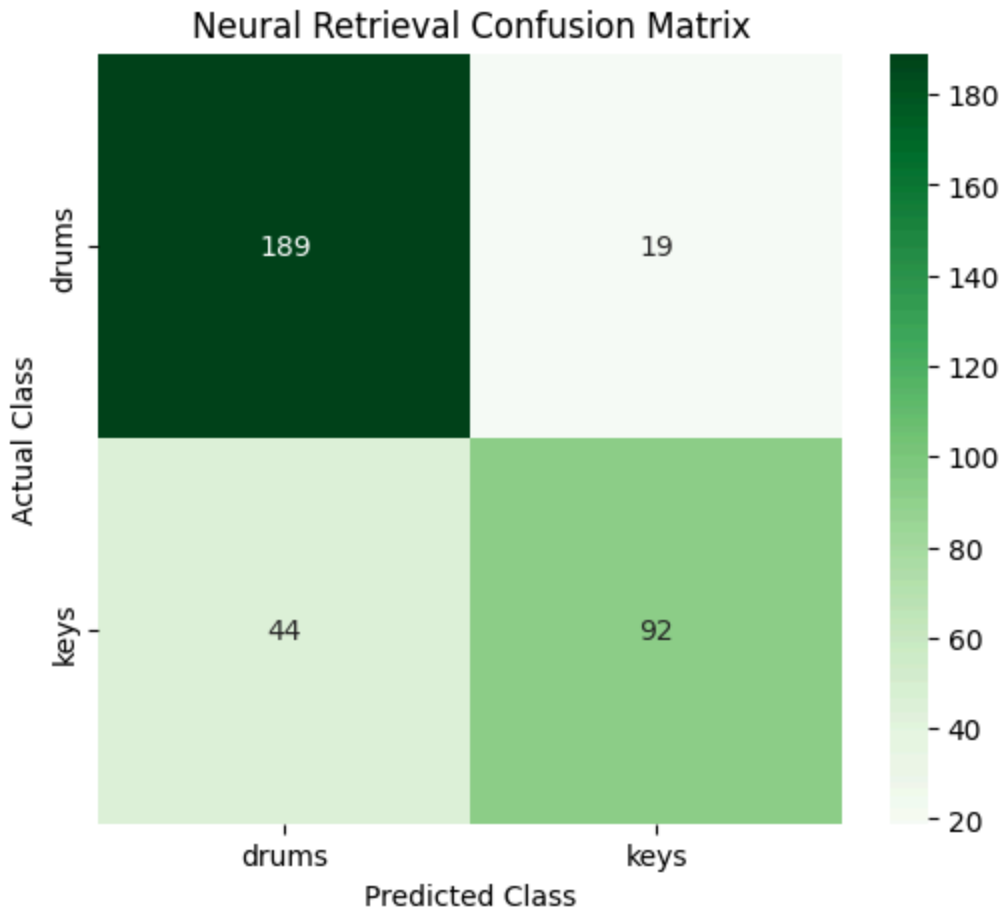
- **Audio Model:** AST (Audio Spectrogram Transformer), trained on AudioSet (general sounds).
- **Text Model:** all-MiniLM-L6-v2, optimized for speed.

Performance Analysis

The baseline model achieved an accuracy of 81.7%. While it successfully identified obvious drum sounds, it struggled with the "Keys" class (Recall: 0.68), often confusing complex tonal textures. The high test loss (~0.11) indicates the model had difficulty finding a stable mapping between the text and audio spaces.

Key Metrics (Model A)

- **Accuracy:** 0.817
- **Precision:** 0.818
- **F1 Score:** 0.813
- **Test Loss:** ~0.11



Model B: The SOTA Approach (Music-Specialized)

Architecture

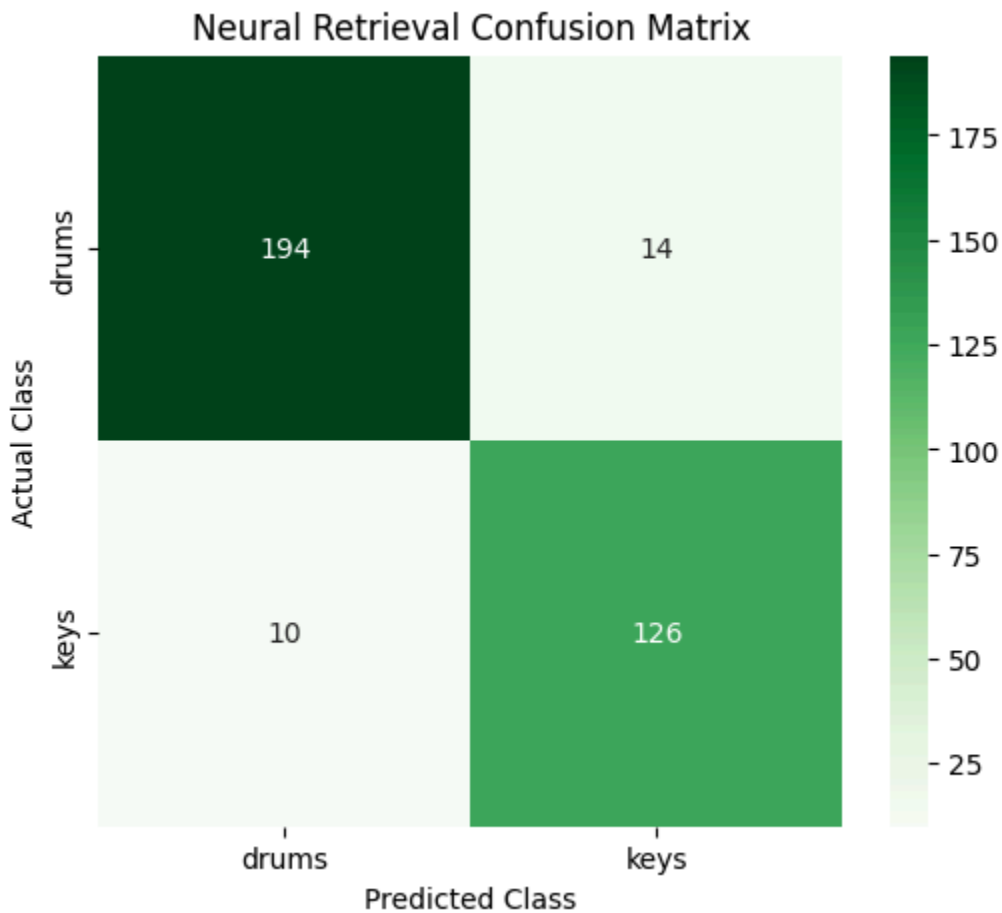
- **Audio Model:** MERT-v1-330M, a State-of-the-Art Music Understanding model trained on 160k hours of music.
- **Text Model:** BGE-Large-en-v1.5, a top-tier semantic embedding model (1024 dimensions).
- **Projector:** Enhanced 2-Layer MLP (1024 -> 2048 -> 4096 -> 1024).

Performance Analysis

The SOTA model delivered a significant performance leap, achieving an accuracy of 93.0%. The use of MERT allowed the system to understand musical nuances, raising the F1 score for the difficult "Keys" class from 0.74 to 0.91. The test loss dropped drastically to 0.015, proving that the larger embeddings provided a much richer landscape for the neural bridge to align the modalities.

Key Metrics (Model B)

- Accuracy: 0.930
- Precision: 0.931
- F1 Score: 0.930
- Test Loss: 0.015



5. Comparative Conclusion

The experiment confirms that while generic audio models (AST) can perform basic retrieval tasks, domain-specific models (MERT) are essential for high-fidelity music retrieval. Model B reduced the error rate by over 60% compared to Model A.

Final Deliverable

The system is fully operational, featuring a ChromaDB backend for scalable search and an interactive frontend for real-time query and playback.

Code Repository

(Link to your GitHub Repository)

